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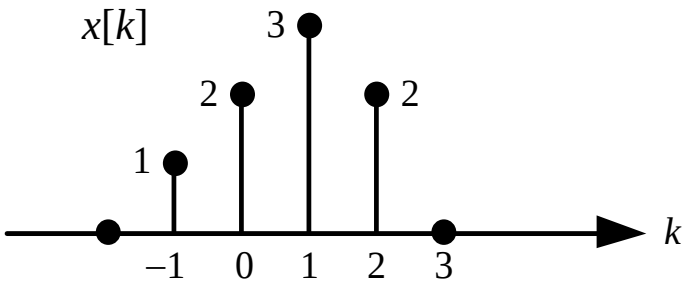
Basic operations of Discrete-Time signals

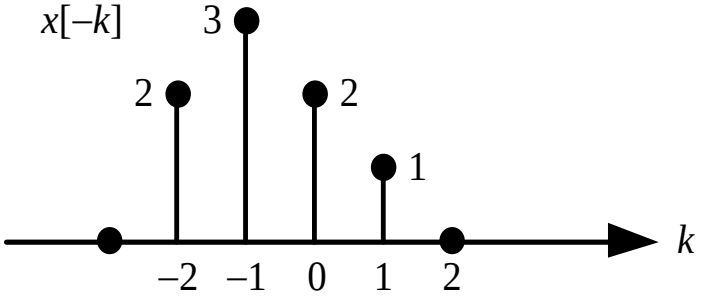
- ※ D-T reversal
- ※ D-T delay
- ※ D-T scaling(decimation & interpolation)
- ※ D-T signal addition
- ※ D-T multiply
- ※ D-T difference
- ※ D-T Summation

D-T reversal

$x[k] \rightarrow x[-k]$

Reversing x[k] along the vertical axis.

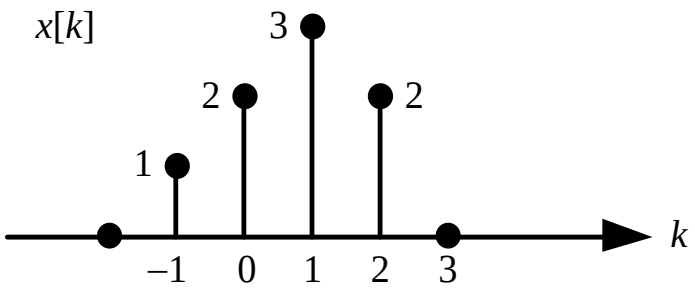


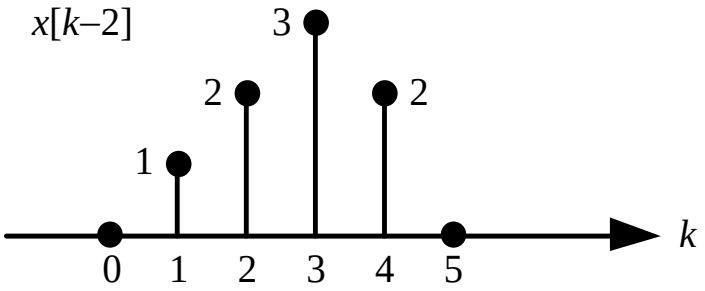


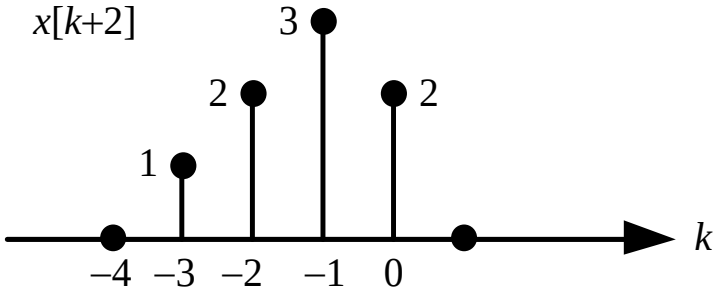
D-T delay

$x[k] \rightarrow x[k \pm n] \quad n>0$

$x[k-n]$ is the right shift of $x[k]$ with n *units*.
$x[k+n]$ is the left shift of $x[k]$ with n *units*.



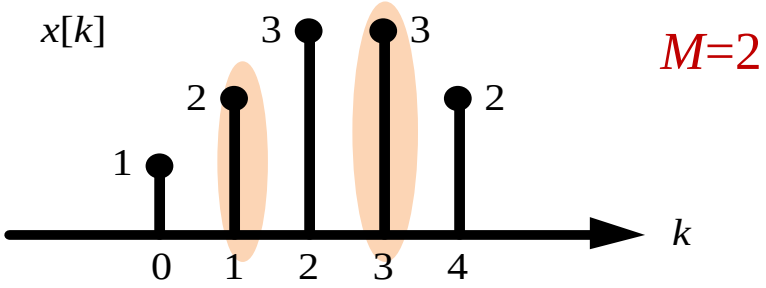




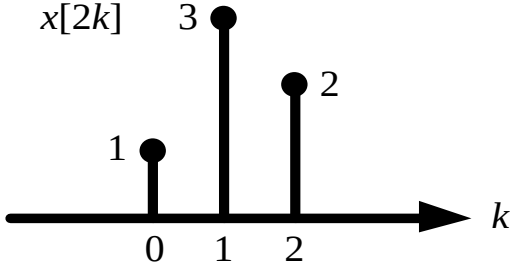
D-T scaling(decimation & interpolation)

$x[k] \rightarrow x[Mk]$ M is positive integer

Decimation $\downarrow M$



$M=2$



The original sequence is extracted one point at every M-1 point

D-T scaling(decimation & interpolation)

Original signal

```
[x,Fs,bits] = wavread('我的祖国');
% Fs=22,050 Hz
```

4x the post-decimation

```
x1=x(1:4:end);
% Fs=22,050/4 Hz
```

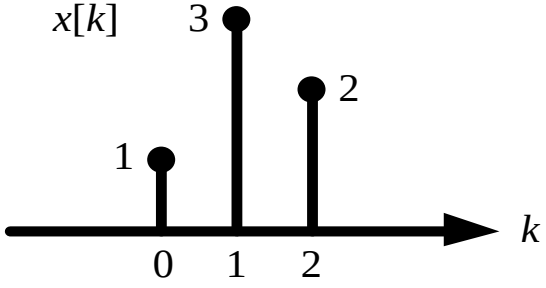
8x the post-decimation

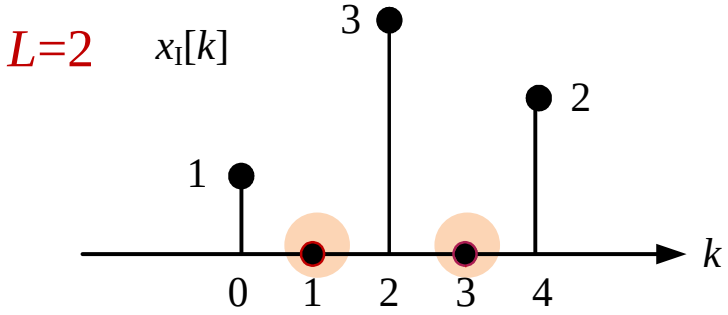
```
x2=x(1:8:end);
% Fs=22,050/8 Hz
```

D-T scaling(decimation & interpolation)

$$x_I[k]=\begin{cases} x[k/L], & k=0,\pm L,\pm 2L,\ldots \\ 0, & \text{otherwise} \end{cases}$$

Interpolation $\uparrow L$





L-1 point is inserted between the points of the original sequence

D-T scaling(decimation & interpolation)

D-T signals are known $x[k]=\{3,\overset{\downarrow}{2},1,0,1,4\}$, draw $x[-3k+2]$.

$$x[-k]=\{4,1,0,1,\overset{\downarrow}{2},3\}$$

Move two units to the right

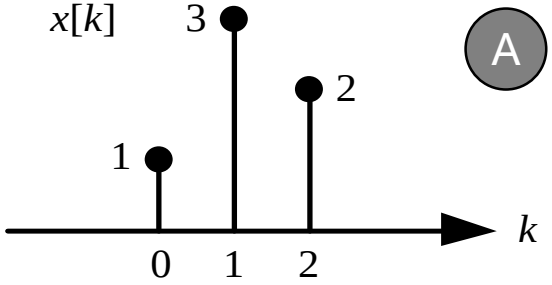
$$x[-k+2]=\{4,1,\overset{\downarrow}{0},1,2,3\}$$

3x extraction

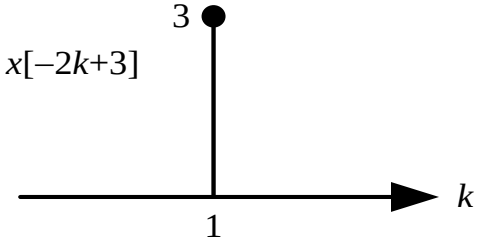
$$x[-3k+2]=\{\overset{\downarrow}{0},3\}$$

Single choice 1 point

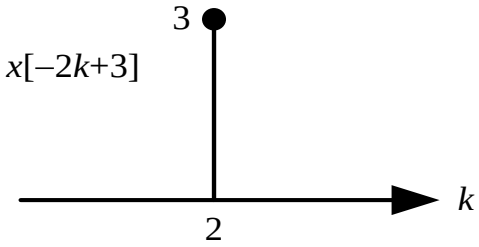
Known signal waveform of $x[k]$ is shown in the following figure, the waveform of $x[-2k+3]$ is



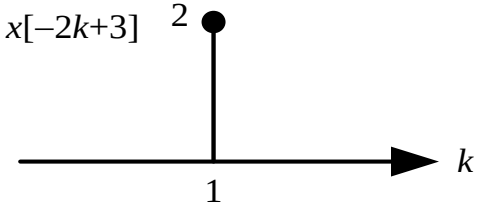
A



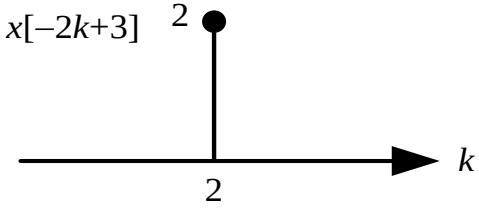
B



C



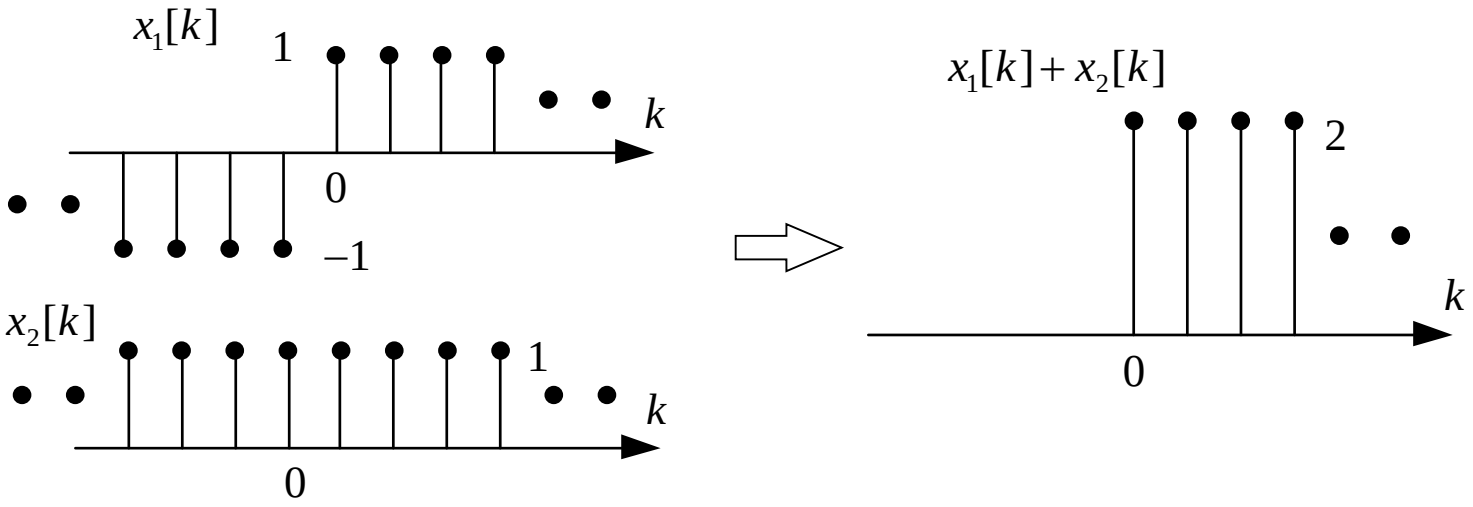
D



Addition

$$y[k] = x_1[k] + x_2[k] + \mathrm{K} + x_n[k]$$

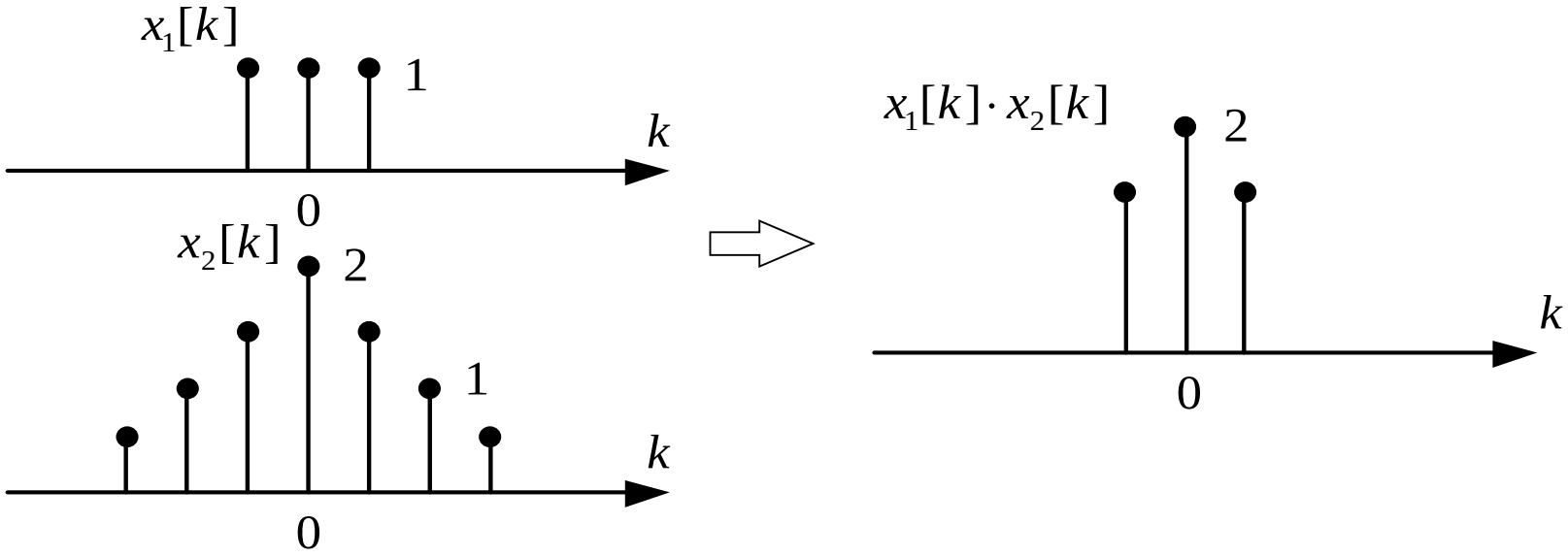
Several discrete sequences add up the corresponding point signal values



Multiplication

$$y[k] = x_1[k] \cdot x_2[k] \cdot \ \mathrm{K} \ \cdot x_n[k]$$

Several discrete sequences multiply the corresponding point signal values



Difference

First-order backward differential $\nabla x[k] = x[k] - x[k-1]$

Second-order backward differential $\nabla^2 x[k] = \nabla\{\nabla x[k]\} = x[k] - 2x[k-1] + x[k-2]$

$\cdots$

N-order backward differential $\nabla^n x[k] = \nabla\{\nabla^{n-1} x[k]\}$

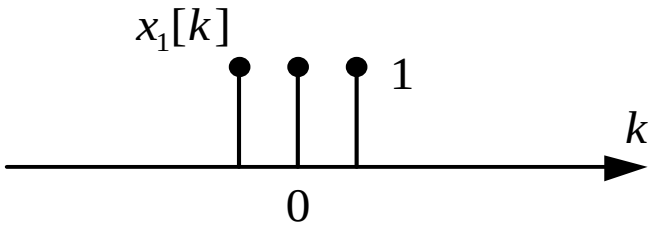
A unit pulse sequence can be expressed as a first-order backward difference of a unit step sequence

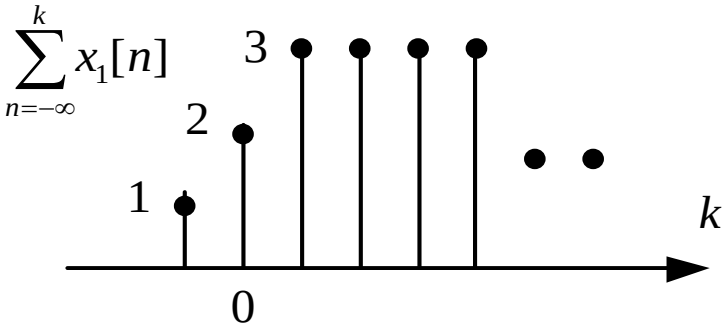
$$\delta[k] = u[k] - u[k-1] = \nabla u[k]$$

First-order forward differential $\Delta x[k] = x[k+1] - x[k]$

Summation

$$y[k] = \sum_{n=-\infty}^{k} x[n]$$





A unit step sequence can be expressed as a summation of a unit pulse sequence

$$u[k] = \sum_{n=-\infty}^{k} \delta[n]$$

Example: A trial basic operation represents the relationship between the following signals

$$\delta(t), u(t), \delta'(t), r(t)$$

$$\delta[k], u[k], r[k]$$

$$\delta(t) = \int_{-\infty}^{t} \delta'(\tau)\mathrm{d}\tau$$

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\mathrm{d}\tau$$

$$r(t) = \int_{-\infty}^{t} u(\tau)\mathrm{d}\tau$$

$$u[k] = \sum_{n=-\infty}^{k} \delta[n]$$

$$r[k] = \sum_{n=-\infty}^{k-1} u[n]$$

Example: A trial basic operation represents the relationship between the following signals

$$\delta(t), u(t), \delta^{'}(t), r(t)$$

$$\delta[k], u[k], r[k]$$

$$\delta^{'}(t) = \delta'(t)$$

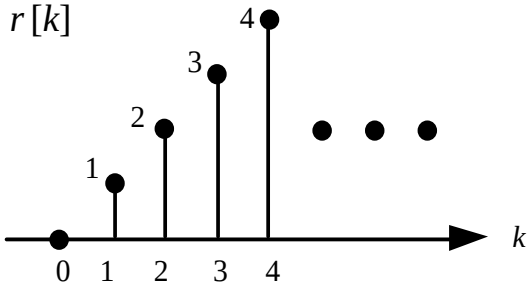
$$\delta(t) = u'(t)$$

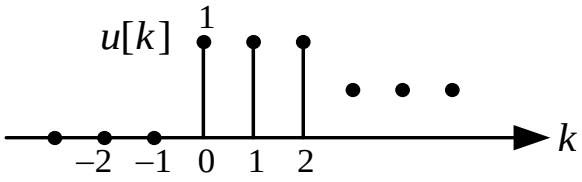
$$u(t) = r'(t)$$

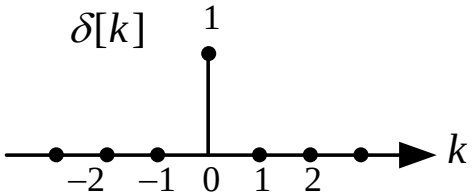
$$\delta[k] = u[k] - u[k-1] = \nabla u[k]$$

$$u[k] = r[k+1] - r[k] = \Delta r[k]$$

brief summary







Differential relationships

$$\delta[k] = \boxed{u[k] - u[k-1]} \qquad u[k] = \boxed{r[k+1] - r[k]}$$

Sum relationships

$$u[k] = \boxed{\sum_{n=-\infty}^{k} \delta[n]} \qquad r[k+1] = \boxed{\sum_{n=-\infty}^{k} u[n]}$$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!